Experiment 15

Career Pathways and Future of Low-Code Development

Aim

To study the career pathways available in low-code development and understand how low-code platforms help individuals build software applications, automate business processes, improve productivity and develop skills for employment and entrepreneurship.

Introduction

Low-code development uses visual tools, drag-and-drop components, predefined templates and minimal programming. Platforms such as Zoho Creator enable applications to be built quickly and business processes to be automated. The Student Admission application demonstrates how these skills can be applied to a real college admissions use case.

Procedure

1. Study the concept of low-code development and commonly used platforms.

2. Identify career roles available in low-code development and the responsibilities of each role.

3. List the technical and non-technical skills required for these roles.

4. Relate each role and skill to the experiments covering the Student Admission form, advanced fields, APIs, roles, security, deployment, Zia and Catalyst.

5. Study future trends including AI integration, business automation and low-code with pro-code.

6. Summarize the findings as career pathways and a skill roadmap.

Career Pathways

1. Low-Code Application Developer — Creates applications using visual platforms, designs forms, reports and workflows. Example: building the Student Admission Enquiry Form and Admission Enquiry Report.

2. Business Process Automation Specialist — Automates repetitive activities such as enquiry acknowledgements, fee calculations, notifications and approvals.

3. Low-Code Solution Architect — Plans application structure, integrations, data flow and business requirements for the admission system.

4. Business Analyst — Understands admission-office requirements and translates them into fields, validation rules, reports and workflows.

5. Integration and Pro-Code Developer — Extends Creator using APIs, JavaScript, Node.js, Python or Catalyst for specialized functions.

6. Low-Code Consultant and Administrator — Manages platform selection, users, permissions, configuration, monitoring and maintenance.

Skills Required

• Platform knowledge: Zoho Creator and its development features.

• Database fundamentals: storing, retrieving and managing enquiry data.

• Basic programming: logic, conditions, loops and Deluge scripting.

• Problem-solving: converting admission requirements into application features.

• Testing and debugging: validating forms, calculations, permissions and integrations.

• Communication skills: understanding requirements and explaining technical solutions.

Future of Low-Code Development

• Integration with Artificial Intelligence for assisted development and data analysis.

• Growth of business automation for approvals, notifications, reporting and applicant management.

• Combination of low-code and pro-code for specialized calculations and integrations.

• Greater focus on security and governance for sensitive applicant information.

• Cloud-based application development across locations and devices.

• Continued need for technical skills in application logic, databases, testing and APIs.

Skill Roadmap and Evidence from This Lab Record

Stage	Skills	Evidence in Student Admission Project
1. Build	Forms, advanced fields, reports	Admission Enquiry Form and Report
2. Automate	Validation, formula, system-generated values	Fee calculation, Auto Number and QR
3. Integrate	APIs, Analytics	Experiment 7
4. Govern	Roles, security, deployment	Experiments 8, 11 and 12
5. Extend	AI, optimization, Catalyst	Experiments 9, 10 and 14

Roles and Matching Project Work

Career Role	Related Work in the Student Admission Project
Low-Code Application Developer	Admission form, advanced fields and reports
Business Process Automation	Fee calculation, QR generation and notifications
Solution Architect	Form, report, API, security and deployment design
Business Analyst	Admission requirements, validations and enquiry workflow
Integration / Pro-Code Developer	REST API and Catalyst fee function
Consultant / Administrator	Roles, permissions, deployment and backups

Result

The career pathways in low-code development, the skills required for them and the future trends of low-code development were studied successfully using the Student Admission project as the practical context.

Conclusion

Low-code development offers career opportunities in application development, workflow automation, testing, consulting and administration. Combining Zoho Creator skills with databases, APIs, AI tools, security and pro-code extensions can help students build practical applications and prepare for entry-level IT roles.